

CODTECH DATA SCIENCE INTERNSHIP TECHNICAL REPORT

CAPSTONE PROJECT: HOUSE PRICE PREDICTION MODEL

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1. Executive Summary

This technical report presents the implementation and validation of an automated machine learning pipeline for predicting residential real estate values. The project leverages the public Kaggle Housing dataset to construct a model estimating house prices based on physical attributes and residential features. Multiple regression algorithms—specifically Linear Regression, Decision Tree Regressor, and Random Forest Regressor—were implemented, optimized, and evaluated. Multiple Linear Regression was selected as the optimal model, attaining an R^2 score of 0.6495. This report outlines the comprehensive framework spanning data preprocessing, exploratory data analysis, feature engineering, and model validation.

2. Project Objective

The capstone project was designed to fulfill the following objectives:

- Design and execute a clean, modular Python machine learning pipeline.
- Conduct detailed Exploratory Data Analysis (EDA) to map feature relationships.
- Preprocess continuous numeric inputs and encode binary/multiclass categorical variables.
- Train and benchmark Linear Regression, Decision Tree, and Random Forest models.
- Evaluate using quantitative metrics: Mean Absolute Error (MAE), RMSE, and R^2 Score.

- Save the champion model to file for downstream application deployment.

3. Dataset Description

The project utilizes a standard tabular real estate dataset comprising 545 observations and 13 architectural/amenity variables. The goal is to predict the price (dependent target variable).

Variable	Type	Description
price	Numeric (Target)	Selling price of the residential property
area	Numeric	Total floor space area in square feet
bedrooms	Discrete	Number of bedrooms
bathrooms	Discrete	Number of bathrooms
stories	Discrete	Number of stories (floors)
mainroad	Categorical	Connecting main road proximity (yes/no)
guestroom	Categorical	Availability of guest rooms (yes/no)
basement	Categorical	Presence of basement level (yes/no)
hotwaterheating	Categorical	Presence of water heating (yes/no)
airconditioning	Categorical	Presence of central cooling (yes/no)
parking	Discrete	Number of parking spaces
prefarea	Categorical	Located in a preferred urban neighborhood
furnishingstatus	Categorical	Furnishing condition (furnished/semi/unfurnished)

4. Methodology

The project follows a standard structured data science pipeline: data acquisition, data cleaning (checking for missing values and duplicates), Exploratory Data Analysis (mapping distributions and correlation matrices), Feature Encoding (mapping binary columns and one-hot encoding furnishings), Train-Test Split (80/20 split), Model Training (benchmarking Linear Regression, Decision Tree, and Random Forest), Evaluation, and Model Selection.

5. Model Building & Evaluation Metrics

To measure prediction accuracy and model generalizability, trained models were tested on the validation split (109 samples) using the following metrics:

- **Mean Absolute Error (MAE):** Calculates the average of the absolute differences between predictions and actual values.
- **Root Mean Squared Error (RMSE):** The square root of the average of squared prediction errors, penalizing larger deviations in the original price unit (\$).
- **R² Score (Coefficient of Determination):** Calculates the proportion of target variance explained by the features.

6. Results & Discussion

Following evaluation, the model performance metrics were compiled below:

Model	MAE (\$)	RMSE (\$)	R ² Score
Linear Regression	979,679.69	1,331,071.42	0.6495
Random Forest Regressor	1,025,289.68	1,401,263.08	0.6115
Decision Tree Regressor	1,243,654.52	1,654,427.54	0.4585

The evaluation shows that Multiple Linear Regression achieved the highest accuracy with an R² score of 0.6495. This indicates a predominantly linear relationship between physical layout (area, bathrooms, stories, parking) and property market valuation. The Random Forest model achieved a close second (R² = 0.6115), but experienced slight overfitting due to the modest sample size (545 rows) which benefits simpler, regularized algorithms. The Decision Tree performed poorly (R² = 0.4585) due to its high variance and split behavior.

7. Conclusions

The capstone project successfully implemented a complete end-to-end Machine Learning prediction pipeline. Physical size, bathroom count, air conditioning, and preferred locations represent the primary economic determinants of home valuation. Simple linear formulations provide robust, generalized predictions for tabular dataset structures. The champion model was serialized and saved as a pickle file.

8. Future Scope

To extend this work, we propose implementing hyperparameter optimization (via grid search) on ensemble estimators, augmenting datasets with geographical attributes (neighborhood coordinates, local school indices), and deploying the serialized model inside a web-based dashboard using Streamlit or Flask APIs.

9. References

1. Scikit-learn Supervised Learning API: <https://scikit-learn.org/stable/>
2. Pandas Data Manipulation Documentation: <https://pandas.pydata.org/>
3. Kaggle House Price Datasets: <https://www.kaggle.com/>